

Changchen Li

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EDUCATION

M.Sc. in Robotics, College of Design and Engineering, National University of Singapore

Aug 2026 – Present

B.Eng. in Engineering Mechanics, School of Aerospace Engineering, Huazhong University of Science and Technology

Sep 2022 – Jun 2026

- *Cumulative GPA:* 4.20/5.00 (87.1/100)
- Rank: 3/19
- Courses: Robotics; Machine Vision and Application; Fundamentals of Computer Programming (C++); Computational Methods; Modeling & Simulation of Engineering Structures; Theoretical Mechanics

HONORS & AWARD

- Outstanding Graduate, Huazhong University of Science and Technology, 2026
- First Prize, Challenge Cup Science & Technology Competition, National level, 2024
- Third Prize, China Undergraduate Mathematical Contest in Modeling, Provincial level, 2024
- Academic Excellence Scholarship 2024–2025
- Academic Excellence Scholarship 2023–2024
- Academic Excellence Scholarship 2022–2023
- Self-improvement Scholarship 2022–2023

INTERNSHIP

SAIC Motor-Smart Chassis Systems.

Chassis Dynamics Department | Jul 2025 - Aug 2025

- Designed a simplified double-wishbone corner module model by analyzing and integrating characteristics of various corner module types
- Utilized topology optimization techniques to enhance the structural performance, resulting in a 51.3% reduction in the maximum stress while increasing the mass by only 0.73% for the corner module
- Simulated worm–gear transmission behaviour in ANSYS to study load-induced deformation

ACADEMIC EXPERIENCE

Undergraduate Training Program for Innovation and Entrepreneurship

National-Level Project | Team leader | Mar 2024 - April 2026

- Led intelligent flight system design and contributed to composite impact analysis
- Led structural design of a foldable composite wing mechanism using Solidworks; filed utility patent (pending)
- Cooperated with teammates on early-stage structural estimation using ABAQUS and Ansys simulations, helping determine wing surface area under estimated velocities

‘Challenge Cup’ Science & Technology Competition

National 1st Prize Winner | Computer Vision Group Leader | Jul 2024 – Nov 2024

- Designed and implemented a target recognition algorithm using a customized ComplexCNN, achieving 99%+ accuracy
- Developed SIFT-based feature matching for region center detection, reaching 95% accuracy
- Reduced the average processing time per image by approximately 67% through image preprocessing and segmentation.

China Undergraduate Mathematical Contest in Modelling(CUMCM)

Provincial Third Prize Winner | Team Leader | Sep 2024

- Modeled dynamics of a snake coiling along a spiral using nodal physics

- Employed gradient descent and binary search methods to solve optimization problems involving spatial positioning

Data Science exchange program in the Imperial College London

Participant | Jan 2024 - Feb 2024

- Preprocessed brain MRI data via registration, resampling, and data augmentation to standardize input for segmentation
- Implemented and trained a 3D U-Net model for glioma region classification, achieving a precision greater than 90% in accuracy

'Robotmaster' Club in HUST

Member of Mechanical Team | Oct 2023 - Jan 2024

- Designed and manufactured carbon fiber guards to protect drone blades from damage
- Designed a special bomb delivery pipe for a drone that could fire rubber bullets during adversarial matches
- Collaborated with teammates to design and build an intelligent vehicle equipped with a gripper capable of autonomously picking up blocks for competition

SKILLS & INTERESTS

Coding Skills: Python (Pytorch, OpenCV), MATLAB, C++

Simulation Softwares: Solidworks, Ansys, COMSOL, ABAQUS

Languages: Mandarin Chinese (native), English (IELTS 7.0)

Interests: Robot Learning; Embodied AI; Multimodal Representation Learning; Large Language Model; Computer Vision